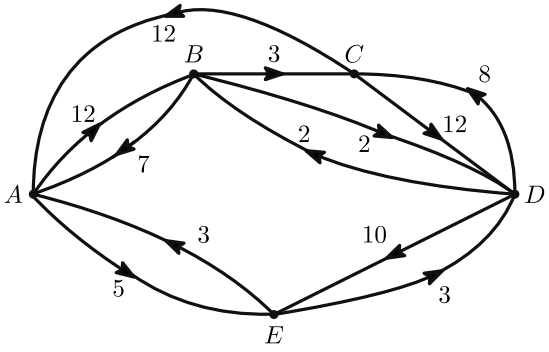
Note. Marks should be awarded suitably to any alternative solutions not given here.

Descriptive Questions

11. Using Dijkstra's algorithm, find shortest paths from A to all the other vertices in the graph given below. (4M)



Ans. The matrix of edge weights is $W = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 12 & \infty & \infty & 5 \\ 7 & 0 & 3 & 2 & \infty \\ 12 & \infty & 0 & 12 & \infty \\ \infty & 2 & 8 & 0 & 10 \\ 3 & \infty & \infty & 3 & 0 \end{bmatrix} \end{array}$.

If the weights matrix is given, it can be awarded $\frac{1}{2}$, but if it is not given, marks should **not** be deducted if the solution is correct – i.e. full marks can be given if the algorithm is applied correctly.

	B	C	D	E	
$A(0)$	12_A	∞_A	∞_A	$\boxed{5_A}$	-1
$E(5)$	12_A	∞_A	$\boxed{8_E}$	$-$	-1
$D(8)$	$\boxed{10_D}$	16_D	$-$	$-$	-1
$B(10)$	$-$	13_B	$-$	$-$	$-\frac{1}{2}$
A	10_D	13_B	8_E	5_A	$-\frac{1}{2}$

Marks should be also be awarded if the correct solution is given in the other format.

---

12. Show that every subgroup of a cyclic group is cyclic. (3M)

Ans. Let $G = \langle g \rangle$, and $H \leq G$. $-\frac{1}{2}$

Let n be the least positive integer such that $g^n \in H$. $-\frac{1}{2}$

Let g^m be an arbitrary element of H. By the division algorithm,

$$m = nq + r, \quad \exists q, r \in \mathbb{Z}, 0 \leq r < n. \qquad -\frac{1}{2}$$

Then,

$$g^r = (g^n)^{-q} \cdot g^m \in H. \qquad -\frac{1}{2}$$

But n is the least positive integer such that $g^n \in H$, and $n > r \geq 0$. Therefore, $r = 0$. $-\frac{1}{2}$

Thus we have $g^m = (g^n)^q$. Hence, $H = \langle g^n \rangle$. $-\frac{1}{2}$

---

13. Draw three non-isomorphic simple graphs on 4 vertices with exactly 3 edges, giving brief reasons. (3M)

Ans. The three non-isomorphic simple $(4,3)$-graphs are shown below.

 — 2

Reasons (any *one* of the following): — 1

- Their degree sequences are different.
- Their diameters are different (3, 2, ∞/undefined).
- The first two are both connected, but one has only two pendant vertices and the other has three (or one has a vertex of degree 3, the other does not). The third is disconnected.

(Any other valid reason).

---

14. Let G be a group and H a subgroup of G. Show that any two left cosets of H in G are either disjoint or identical. Also illustrate this result with an example. (3M)

Ans. Let xH and yH be two left cosets of H in G, and suppose that $xH \cap yH \neq \emptyset$. — $\frac{1}{2}$

Then there exists an element $xh_1 = yh_2 \in xH \cap yH$, for some $h_1, h_2 \in H$. Hence, $x = yh_2h_1^{-1}$ — $\frac{1}{2}$

Now, for all $xh \in xH$, $xh = yh_2h_1^{-1}h \in yH$, hence $xH \subseteq yH$. — $\frac{1}{2}$

Similarly, $yH \subseteq xH$. Therefore, $xH = yH$. — $\frac{1}{2}$

Any example of a group G and a subgroup H, giving all left cosets of H in G. — 1

---

15. Show that a graph is a tree if and only if, between every pair of vertices, there exists a unique path. Draw a tree on 5 vertices. (3M)

Ans. Suppse that G is a tree, and let u and v be vertices of G. Since G is connected, there is a path, say P_1, joining u and v. Suppose, if possible, that there exists another path P_2 from u to v. — $\frac{1}{2}$

When traversing P_1 from u to v, let x be the first vertex whose successor on P_1 is not on P_2, and let y be the next vertex common to both P_1 and P_2. — $\frac{1}{2}$

Then the segments of P_1 and P_2 between x and y form a cycle in G, which is impossible. Hence, there is at most one path between any two vertices of G. — $\frac{1}{2}$

Conversely, suppose that G is a graph in which there is a unique path joining any two vertices. Clearly, G is connected. — $\frac{1}{2}$

Suppose, if possible that $v_1, v_2, \ldots, v_n$ is a cycle in G. Then the path $v_1, v_2, \ldots, v_n$ and the path v_1, v_n are two different paths from v_1 to v_n, which is impossible. Hence, G is acyclic, and is a tree. — $\frac{1}{2}$

Any tree of order 5. — $\frac{1}{2}$

---

16. Show that a group G is Abelian if and only if, for all $a, b \in G$, $(ab)^2 = a^2b^2$. Give an example of an Abelian group. (3M)

Ans. If G is Abelian, then $(ab)^2 = abab = aabb = a^2b^2$. -1

Conversely,

$$(ab)^2 = a^2b^2 \implies$$
$$abab = aabb \implies \quad -\tfrac{1}{2}$$
$$ba = ab \quad \text{(left and right cancellation).} \quad -1$$

Any example of an Abelian group. $-\frac{1}{2}$

---

17. Let G be a self-complementary graph. Show that the number of vertices in G is of the form $4k$ or $4k+1$. (2M)

Ans. Let n be the order and m the size of G. Then $\overline{G}$ has size $\binom{n}{2} - m$. $-\frac{1}{2}$

Since $G \cong \overline{G}$, $m = \binom{n}{2} - m$. $-\frac{1}{2}$

Hence, $m = \frac{n(n-1)}{4}$. $-\frac{1}{2}$

Since m is an integer, this implies that $n = 4k$ or $n-1 = 4k$, i.e. $n = 4k+1$, for some integer k. $-\frac{1}{2}$

---

18. Let G be a finite group and let H and K be subgroups with relatively prime order. Then show that $H \cap K = \{e\}$. (2M)

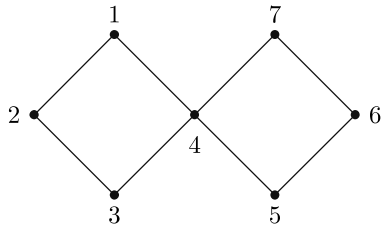
Ans. Since $H \cap K \leq H, K$, $|H \cap K$ divides both $|H|$ and $|K|$. -1

As $|H|$ and $|K|$ are relatively prime, this implies that $|H \cap K| = 1$. $-\frac{1}{2}$

The only subgroup of order 1 is $\{e\}$. $-\frac{1}{2}$

---

19. Show that the graph given below is bipartite, by listing the vertices in each of the partite sets V_1 and V_2. State, with justification, whether this graph is Eulerian. (2M)



Ans. $V_1 = \{1, 3, 5, 7\}$ and $V_2 = \{2, 4, 6\}$ (or vice-versa). -1

The graph is connected and all its vertex-degrees are even. Hence it is Eulerian. -1